AN ORDINAL-PRIME CMZ DEFORMATION CRITERION FOR THE RIEMANN HYPOTHESIS (V4)

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Abstract. We give a self-contained arithmetic-to-dynamical bridge between the Riemann hypothesis and a two-channel logarithmic stretching observable motivated by the multiscale mechanisms of Córdoba, Martínez-Zoroa, and Zheng. The arithmetic input is the sign-specific ordinal-prime transform S, its summatory function SS, and the Dirichlet series α. An exact finite Abel identity, including its unfinished-pair boundary term, identifies the residual Dirichlet polynomial $\delta_Q(\sigma)$ with the logarithm of the relative deformation of two positive CMZ-type channels. We prove that the Riemann hypothesis is equivalent to tail stability of this relative deformation for every real $\sigma > 3/2$. We also record the domains of absolute convergence, all justified interchanges, the meromorphic blocks of α, and the singularities that cannot be transferred to the full series without additional continuation. The equivalence is rigorous; a realisation by the published Euler gluing scheme is stated separately as an open PDE problem and is not asserted here. Three concrete attempts to feed the ordinal-prime data into a genuine forced Euler or IPM tower are examined: each is shown either to require an extension of the cited theorems that is not on record, or to encode the arithmetic input without constraining it. A bounded-ratio generalisation of the unforced layered self-interaction bound of Córdoba, Martínez-Zoroa, and Zheng is proved along the way. The precise sense in which completing any such realisation would still not decide the Riemann hypothesis is stated as Warning 9.23. Two further candidates are then excluded outright rather than left unproved: reading the data off a single tower's amplitude ratio conflicts with an already-proved sign-definite lemma, and reading it off a two-tower comparison of activated packets conflicts, for large arithmetic index q, with an unconditional quadratic-summability bound proved here for the target sequence. The one candidate excluded by neither obstruction — reading the data off a residual term the construction already produces, rather than off an amplified response — is developed here. The open ordinal pairs encoded by SS are placed in distinct prime harmonics. Their formal zero-mode energy is exactly $|M(s_q)|/2$. A global phase average makes that extraction intrinsic and, after recombining the first CMZ correction with the exact multiplier remainder, gives a coercive weak self-interaction proportional to $|M(s_q)|$. We prove this finite-stage statement with its harmonic-capacity, support, boundary, and multiplier hypotheses. We then identify the two estimates still needed for an RH proof: preventing cancellation by the later CMZ corrections and bounding the resulting *normalised* residual by $O_\eta(q^{1/2+\eta})$ through a fluid estimate independent of Mertens cancellation. Neither estimate follows from the cited papers, so the bridge is rigorous but the final RH implication remains explicitly [OPEN].

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Remark 0.1 (Mathematical status and provenance). Statements marked [LIT] are imported from the cited literature. Statements marked [DED] are proved in this article from the stated definitions. Statements marked [ADAPT] adapt a published mechanism to a different object. A statement marked [OPEN] is not proved. The exact ordinal identities and the CMZ–RH equivalence are [DED]. The scalar stretching law is an [ADAPT] of the CMZ amplitude mechanism. No theorem in this article claims that the existing Euler, Boussinesq, or IPM constructions supply the missing arithmetic tail estimate.

1. Introduction

For $s = \sigma + it$ with $\sigma > 1$, the Riemann zeta function is defined by the absolutely convergent Dirichlet series and Euler product

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p} \left(1 - p^{-s}\right)^{-1}. \tag{1.1}$$

It extends meromorphically to $\mathbb{C}$, has a simple pole at $s = 1$, and satisfies a functional equation relating s and $1 - s$; see [3, 1]. Its nontrivial zeros lie in $0 < \Re s < 1$. The Riemann hypothesis is the assertion

$$\mathrm{RH}: \qquad \zeta(\rho) = 0, \quad 0 < \Re\rho < 1 \quad \Longrightarrow \quad \Re\rho = \frac{1}{2}. \tag{1.2}$$

The bridge developed here begins with four classical arithmetic functions. The Möbius function μ, the Liouville function λ, and the von Mangoldt function Λ are

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n \text{ is a product of } r \text{ distinct primes}, \\ 0, & p^2 \mid n \text{ for some prime } p, \end{cases} \tag{1.3}$$

$$\lambda(n) = (-1)^{\Omega(n)}, \tag{1.4}$$

$$\Lambda(n) = \begin{cases} \log p, & n = p^k \text{ with } p \text{ prime and } k \geq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{1.5}$$

Here $\Omega(n)$ counts prime factors with multiplicity. Absolute convergence and multiplication of Euler factors give, for $\sigma > 1$,

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}, \tag{1.6}$$

$$\frac{\zeta(2s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s}, \tag{1.7}$$

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}. \tag{1.8}$$

The first identity makes cancellation in μ directly sensitive to zeros of ζ. The second and third identities explain why Liouville signs and prime-power weights are natural comparison objects. The construction below uses μ itself; λ and Λ are introductory reference points, not substituted coefficients.

Write

$$M(x) = \sum_{n \le x} \mu(n). \tag{1.9}$$

The classical Littlewood criterion states

$$\text{RH} \iff M(x) = O_\eta\Big(x^{1/2+\eta}\Big) \quad \text{for every } \eta > 0. \tag{1.10}$$

We use this imported theorem in its stated pointwise form [2, 3]. The objective of this article is not to replace (1.10) by an unproved estimate under a new name. It is to identify exactly how an ordinal-prime observable enters a CMZ logarithmic stretching law and to state the resulting equivalence without discarding cut-off or analytic-domain information.

2. The ordinal-prime transform

2.1. **Counters and transformed values.** Let p_j be the jth prime and put $p_0 = 0$. For real $x \ge 1$, define

$$N_+(x) = \#\{n \le x : \mu(n) = 1\}, \tag{2.1}$$

$$N_-(x) = \#\{n \le x : \mu(n) = -1\}, \tag{2.2}$$

$$Q(x) = N_+(x) + N_-(x), \tag{2.3}$$

$$M(x) = N_+(x) - N_-(x). \tag{2.4}$$

The last definition agrees with (1.9). In particular,

$$N_\pm(x) = \frac{Q(x) \pm M(x)}{2}. \tag{2.5}$$

Definition 2.1 (Ordinal-prime transform)**.** The sign-specific ordinal-prime transform and its summatory function are

$$S(n) = \begin{cases} p_{N_+(n)}, & \mu(n) = 1, \\ -p_{N_-(n)}, & \mu(n) = -1, \\ 1, & \mu(n) = 0, \end{cases} \tag{2.6}$$

$$SS(x) = \sum_{n \le x} S(n). \tag{2.7}$$

Thus the first positive Möbius value receives p_1, the second receives p_2, and so on; the negative occurrences receive the negatives of the same prime list on their own clock. The value 1 at nonsquarefree integers is part of the definition and must be retained.

Put

$$\mathcal{P}(m) = \sum_{j=1}^{m} p_j, \qquad \mathcal{P}(0) = 0. \tag{2.8}$$

Proposition 2.2 (Closed form for SS)**.** *For every real $x \geq 1$,*

$$SS(x) = \lfloor x \rfloor - Q(x) + \mathcal{P}\big(N_+(x)\big) - \mathcal{P}\big(N_-(x)\big). \tag{2.9}$$

Consequently, with

$$T(x) = SS(x) - \lfloor x \rfloor + Q(x), \tag{2.10}$$

one has

$$T(x) = \mathcal{P}\big(N_+(x)\big) - \mathcal{P}\big(N_-(x)\big). \tag{2.11}$$

Proof. There are $\lfloor x \rfloor - Q(x)$ nonsquarefree integers up to x, each contributing 1. The positive squarefree occurrences contribute the prime prefix $p_1 + \cdots + p_{N_+(x)}$, and the negative occurrences contribute the negative prime prefix $-(p_1 + \cdots + p_{N_-(x)})$. Their sum is (2.9); subtracting the zero row gives (2.11). □

2.2. **Ordinal and prime-displacement parts.** For a squarefree n, let $j(n)$ denote its occurrence number inside its own sign class. Define

$$w_j = p_j - j, \qquad D_w(m) = \sum_{j=1}^{m} w_j, \qquad D_w(0) = 0. \tag{2.12}$$

Lemma 2.3 (Exact ordinal coefficient sum)**.** *For every real $x \geq 1$,*

$$\sum_{\substack{n \leq x \\ \mu(n) \neq 0}} \mu(n) j(n) = \frac{M(x)(Q(x)+1)}{2}. \tag{2.13}$$

Proof. The positive occurrences contribute $1 + \cdots + N_+$ and the negative occurrences contribute $-(1 + \cdots + N_-)$. Therefore

$$\sum_{\substack{n \leq x \\ \mu(n) \neq 0}} \mu(n) j(n) = \frac{N_+(N_+ + 1) - N_-(N_- + 1)}{2} \tag{2.14}$$

$$= \frac{(N_+ - N_-)(N_+ + N_- + 1)}{2} \tag{2.15}$$

$$= \frac{M(x)(Q(x)+1)}{2}. \tag{2.16}$$

□

Definition 2.4 (Prime-displacement partial sum)**.** Set

$$W(x) = \sum_{\substack{n \leq x \\ \mu(n) \neq 0}} \mu(n) w_{j(n)}. \tag{2.17}$$

Proposition 2.5 (Exact recovery of W from SS)**.** *For every real $x \geq 1$,*

$$W(x) = D_w\big(N_+(x)\big) - D_w\big(N_-(x)\big) \tag{2.18}$$

$$= SS(x) - \lfloor x \rfloor + Q(x) - \frac{M(x)(Q(x)+1)}{2}. \tag{2.19}$$

Proof. The first equality follows by summing w_j on the two occurrence clocks. Since $p_j = j + w_j$, the signed-prime contribution $T(x)$ is the sum of the ordinal contribution in Lemma 2.3 and $W(x)$. Use (2.10) and rearrange. □

Proposition 2.6 (Asymptotic amplification)**.** *Let $c_0 = 3/\pi^2$. As $x \to \infty$,*

$$W(x) = c_0 x \log x\, M(x)\big(1 + o(1)\big), \tag{2.20}$$

with equality of both sides when $M(x) = 0$ understood before taking the relative error.

Proof. The prime number theorem gives $p_j \sim j \log j$ and $M(x) = o(x)$, while elementary squarefree counting gives

$$Q(x) = \frac{6}{\pi^2} x + O(\sqrt{x}) = 2c_0 x + O(\sqrt{x}). \tag{2.21}$$

Hence both $N_+(x)$ and $N_-(x)$ are $c_0 x(1 + o(1))$. If $M(x) > 0$, (2.18) is the block sum of w_j over $N_-(x) < j \le N_+(x)$; if $M(x) < 0$, it is the negative of the reversed block. Every index in that block is $c_0 x(1 + o(1))$, uniformly, and

$$w_j = p_j - j = c_0 x \log x(1 + o(1)) \tag{2.22}$$

uniformly on the block. There are $|M(x)|$ terms and their common orientation is $\operatorname{sgn} M(x)$, proving (2.20). □

3. The Dirichlet series α and its exact decomposition

Definition 3.1 (The series α and δ)**.** Define

$$\alpha(s) = \sum_{n=1}^{\infty} \frac{S(n)}{n^s}, \tag{3.1}$$

$$\delta(s) = \sum_{\substack{n \ge 1 \\ \mu(n) \ne 0}} \frac{\mu(n) w_{j(n)}}{n^s}. \tag{3.2}$$

The analytic function $\alpha(s)$ is not the ordinal position α_j introduced later.

Since $p_j \ll j \log(2j)$ and $j(n) \le n$, both series in Definition 3.1 converge absolutely in $\Re s > 2$. This is the unconditional half-plane in which termwise regrouping will first be used.

Proposition 3.2 (Four-block decomposition)**.** *For $\Re s > 2$,*

$$\alpha(s) = A(s) + B(s) + C(s) + \delta(s), \tag{3.3}$$

where

$$A(s) = \zeta(s) - \frac{\zeta(s)}{\zeta(2s)}, \tag{3.4}$$

$$B(s) = c_0 \frac{s}{(s-1)\zeta(s-1)}, \tag{3.5}$$

$$C(s) = \frac{s}{2} \int_1^{\infty} M(x)\big(Q(x) + 1 - 2c_0 x\big) x^{-s-1}\, dx. \tag{3.6}$$

The functions A and B are meromorphic on $\mathbb{C}$. The integral defining C converges locally uniformly and defines a holomorphic function in $\Re s > 3/2$.

Proof. The nonsquarefree row of S equals

$$\sum_{\mu(n)=0} \frac{1}{n^s} = \sum_{n \ge 1} \frac{1}{n^s} - \sum_{n \ge 1} \frac{\mu(n)^2}{n^s} \tag{3.7}$$

$$= \zeta(s) - \frac{\zeta(s)}{\zeta(2s)} = A(s), \tag{3.8}$$

where the second equality uses the Euler product for the squarefree indicator. All series are absolutely convergent for $\Re s > 2$, in fact for this step $\Re s > 1$ suffices.

For squarefree n, the identity $p_{j(n)} = j(n) + w_{j(n)}$ gives

$$S(n) = \mu(n)j(n) + \mu(n)w_{j(n)}. \tag{3.9}$$

The second term produces δ. Let

$$J(x) = \sum_{\substack{n\le x\\ \mu(n)\neq 0}} \mu(n)j(n) = \frac{M(x)(Q(x)+1)}{2}. \tag{3.10}$$

Finite Abel summation followed by passage to infinity yields

$$\sum_{\substack{n\ge 1\\ \mu(n)\neq 0}} \frac{\mu(n)j(n)}{n^s} = s\int_1^\infty J(x)x^{-s-1}\,dx, \tag{3.11}$$

because $J(x) = O(x^2)$ and $J(X)X^{-s} \to 0$ for $\Re s > 2$. Split $Q(x)+1$ as $2c_0x + (Q(x)+1-2c_0x)$. The remainder is exactly $C(s)$. For the main term, put $z = s-1$ and use

$$\int_1^\infty M(x)x^{-z-1}\,dx = \frac{1}{z\zeta(z)}, \qquad \Re z > 1, \tag{3.12}$$

which follows by Abel summation from (1.6). The main term is therefore $B(s)$.

Finally, (2.21) and the unconditional bound $|M(x)| \le x$ give

$$\left|M(x)(Q(x)+1-2c_0x)x^{-s-1}\right| \ll x^{1/2-\Re s}. \tag{3.13}$$

The majorant is integrable precisely when $\Re s > 3/2$. On every compact subset of that half-plane it is uniform, so dominated convergence for holomorphic parameter integrals proves holomorphy of C there. □

3.1. **Singularities and the noncancellation warning.**

Proposition 3.3 (Singularities of the explicit shifted block)**.** *If ρ is a nontrivial zero of ζ of multiplicity m, then B has a pole of order m at $s = 1+\rho$. For a simple zero,*

$$\operatorname*{Res}_{s=1+\rho} B(s) = c_0\frac{1+\rho}{\rho\zeta'(\rho)}. \tag{3.14}$$

The function B also has a simple pole at $s = 1$ with residue $-2c_0$, poles at $s = 1-2k$ arising from the trivial zeros, and a zero rather than a pole at $s = 2$.

Proof. At a zero ρ, factor $\zeta(s-1)$ locally and note that $c_0 s/(s-1)$ is finite and nonzero at $1+\rho$. Formula (3.14) follows from the linear factor at a simple zero. At $s = 1$, use $\zeta(0) = -1/2$. At $s = 2$, the pole of $\zeta(s-1)$ makes its reciprocal vanish. The trivial zeros of $\zeta(s-1)$ occur at $s = 1-2k$. □

Warning 3.4 (No automatic poles for the full α). Proposition 3.3 locates poles of B, not of the full α. Identity (3.3) is initially established only in $\Re s > 2$. A pole of B cannot be assigned to α outside that common domain without first continuing the other terms and proving noncancellation. In particular, neither δ nor C may be silently absorbed into an unspecified holomorphic remainder.

4. Squarefree time, ordinal pairs, and finite boundaries

Let $s_1 < s_2 < \cdots$ enumerate the squarefree positive integers, and define

$$\varepsilon_q = \mu(s_q), \qquad R(q) = \sum_{r=1}^{q}\varepsilon_r = M(s_q), \qquad R(0) = 0. \tag{4.1}$$

The occurrence counters at squarefree time are

$$u_q = \frac{q+R(q)}{2}, qquadv_q = \frac{q-R(q)}{2}. \tag{4.2}$$

Let α_j and β_j denote the squarefree-time positions of the jth positive and negative signs, respectively, and put

$$a_j = s_{\alpha_j}, \qquad b_j = s_{\beta_j}, \qquad d_j = \beta_j - \alpha_j. \tag{4.3}$$

Proposition 4.1 (Pair representation)**.** *For $\Re s > 2$,*

$$\delta(s) = \sum_{j=1}^{\infty} w_j \left(a_j^{-s} - b_j^{-s} \right). \tag{4.4}$$

Moreover, for every complex s and every j,

$$a_j^{-s} - b_j^{-s} = s \int_{a_j}^{b_j} x^{-s-1}\, dx, \tag{4.5}$$

where the orientation of the integral is retained.

Proof. Absolute convergence in $\Re s > 2$ permits matching the jth positive and negative occurrences. Their residual coefficients are w_j and $-w_j$, which gives (4.4). The derivative of x^{-s} is $-sx^{-s-1}$, proving (4.5); the formula also holds at $s = 0$ by direct evaluation. □

Define the coefficient at squarefree time and its cumulative sum by

$$c_q = \varepsilon_q w_{j(s_q)}, \tag{4.6}$$

$$U(q) = \sum_{r=1}^{q} c_r = D_w(u_q) - D_w(v_q). \tag{4.7}$$

Proposition 4.2 (Open-pair block and SS readout)**.** *For every $q \geq 1$,*

$$U(q) = \operatorname{sgn} R(q) \sum_{j=\min(u_q,v_q)+1}^{\max(u_q,v_q)} w_j, \tag{4.8}$$

$$U(q) = SS(s_q) - s_q + q - \frac{q+1}{2} R(q). \tag{4.9}$$

An empty sum in (4.8) *is zero.*

Proof. The difference $D_w(u_q) - D_w(v_q)$ is the oriented block in (4.8). It consists exactly of ordinal pairs with one endpoint at or before cut q and the other endpoint after it. Formula (4.9) is (2.19) evaluated at $x = s_q$, using $Q(s_q) = q$ and $M(s_q) = R(q)$. □

Lemma 4.3 (Finite Abel identity with the endpoint term)**.** *For $Q \geq 1$ and every $s \in \mathbb{C}$, let*

$$\delta_Q(s) = \sum_{q=1}^{Q} c_q s_q^{-s}. \tag{4.10}$$

Then

$$\delta_Q(s) = U(Q) s_Q^{-s} + \sum_{q=1}^{Q-1} U(q) \left(s_q^{-s} - s_{q+1}^{-s} \right). \tag{4.11}$$

Proof. Since $c_q = U(q) - U(q-1)$, with $U(0) = 0$, finite summation gives

$$\sum_{q=1}^{Q} c_q s_q^{-s} = \sum_{q=1}^{Q} \big(U(q) - U(q-1)\big) s_q^{-s} \tag{4.12}$$

$$= U(Q) s_Q^{-s} + \sum_{q=1}^{Q-1} U(q) (s_q^{-s} - s_{q+1}^{-s}). \tag{4.13}$$

No infinite interchange occurs in this proof. □

Remark 4.4 (Meaning of the boundary). The term $U(Q)s_Q^{-s}$ is the contribution of all ordinal pairs still open at the final cut. Removing it replaces the original order by a completed-pair grouping. Such a replacement is not legitimate in the RH transfer unless the boundary is independently shown to vanish.

For $\Re s > 2$, the bound $|U(q)| \ll s_q^2 \log(2s_q)$ makes the infinite cut representation absolutely convergent, and Lemma 4.3 then yields

$$\delta(s) = \sum_{q=1}^{\infty} U(q) \left(s_q^{-s} - s_{q+1}^{-s} \right). \tag{4.14}$$

This passage is not used to define the finite CMZ observable; the finite identity (4.11) is the primary formula.

5. The CMZ multiscale mechanism: a proof-architecture summary

5.1. **Euler vorticity and hyperbolic stretching.** The forced incompressible Euler equations and their vorticity formulation are

$$\partial_t u + (u \cdot \nabla)u + \nabla p = f, \qquad \nabla \cdot u = 0, \tag{5.1}$$

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nabla \times f, \qquad \omega = \nabla \times u. \tag{5.2}$$

The CMZ constructions arrange an older vorticity field whose velocity is approximately linear near the origin. A younger, much smaller oscillatory packet is transported and stretched by this hyperbolic linearisation [5, 6].

In the forced non-axisymmetric construction, a parent scale N and child scale L are related by

$$L^{1/2} = exp\left(\int_{1-N^{-\varepsilon/2}}^{1} \partial_1 u_{N,1}(0,t)\, dt \right), \tag{5.3}$$

under a positive strain bound of size N^{ε}. Writing

$$K_N(t) = \exp\left(\int_t^1 \partial_1 u_{N,1}(0,s)\, ds \right), \qquad b(t) = \frac{\log K_N(t)}{\log L}, \tag{5.4}$$

the transported child packet has anisotropic widths of orders

$$L^{-1/2-b+\varepsilon}, \qquad L^{-1/2-\varepsilon+b}, \qquad L^{-1/2}, \tag{5.5}$$

and oscillation frequency of order L.

5.2. **Parametrix, cancellation, and gluing.** The proof architecture in [5] has the following linked stages.

C1. A local Biot–Savart parametrix approximates the two leading velocity components generated by the packet. For spatial derivatives of order $r = 0, 1$, the quoted error has the form

$$\|\widetilde{u}_j - u_j\|_{C^r} \le C_\kappa L^{2\varepsilon+r-3/2+\kappa}. \tag{5.6}$$

C2. The leading pieces of $\widetilde{u}_2\partial_2\omega_1$ and $\widetilde{u}_3\partial_3\omega_1$ cancel. The proof printed after the stated lemma safely yields

$$\|u(\omega) \cdot \nabla\omega\|_{C^\vartheta} + \|\omega \cdot \nabla u(\omega)\|_{C^\vartheta} \lesssim L^{4\varepsilon-1/2+\vartheta}. \tag{5.7}$$

The exponent 4ε in (5.7), rather than the 3ε appearing in the lemma statement, is the exponent obtained after the paper chooses its auxiliary loss parameter equal to ε.

C3. Oscillation and spatial separation make the child's velocity decay faster than any prescribed power outside its anisotropic support. Odd parity and Taylor expansion control interactions with the parent field.

C4. Replacing the parent velocity by its local linear model produces a controlled forcing error. With $\vartheta = 1/2 - 6\varepsilon$, the safe power in (5.7) becomes $L^{-2\varepsilon}$.

C5. The induction alternates the expanding direction and adds a child whose strain at the origin is comparable to L^{ε}, while its forcing increment is bounded by

$$\|F_{\mathrm{child}}\|_{C^{1/2-6\varepsilon}} \lesssim L^{-2\varepsilon}. \tag{5.8}$$

Choosing scales sufficiently separated makes the forcing increments summable.

These steps summarise the actual proof mechanism; they do not assert that an arbitrary prescribed time-dependent strain can be realised with constants independent of its magnitude.

5.3. **The separated-layer amplitude system.** In the unforced separated-layer construction of [6], the height $x_k(t) > 0$ of the kth packet is governed, up to controlled geometric errors, by

$$\dot{x}_k(t) = x_k(t) \sum_{j<k} a_j(t) x_j(t), \tag{5.9}$$

where $a_j(t)$ is close to 1 and, importantly, does not depend on the child index k. In the exact model $a_j = 1$, writing $y_k = \log x_k$ gives

$$\dot{y}_k(t) = \sum_{j<k} x_j(t). \tag{5.10}$$

For terminal data $x_k(0) = A^k$, $A > 1$, the model admits a uniform bound on the time integral of each amplitude over a fixed backward interval:

$$\int_{-a}^{0} x_k(t)\, dt \le a, \qquad a = A(1 - e^{-a}). \tag{5.11}$$

This positivity is structural. Encoding μ by allowing signed amplitudes would destroy the hypothesis behind (5.11). The two-channel construction in Section 6 avoids that problem by keeping both channels positive.

5.4. **Later multiscale developments.** The 2026 IPM manuscript [7] transports oscillatory packets through nested Fourier cutoffs and proves joint space–time smoothness of the force on the torus. The 2026 Boussinesq manuscript [8] uses an exact growing wave, compactly supported corrections, lifetime derivative estimates, and summable local corrections to obtain smooth forcing in both equations. These works confirm the flexibility of the multiscale programme. Their theorems concern different PDEs and do not imply an arithmetic cancellation estimate for $U(q)$.

6. The positive two-channel CMZ deformation

The construction in this section is an [ADAPT] of the logarithmic stretching law (5.10). It is an exact finite-dimensional dynamical encoding, not yet an Euler solution.

Fix a real $\sigma > 0$ and a cutoff $Q \ge 1$. Define positive clock lengths

$$\ell_q^{(Q)}(\sigma) = \begin{cases} s_q^{-\sigma} - s_{q+1}^{-\sigma}, & 1 \le q < Q, \\ s_Q^{-\sigma}, & q = Q. \end{cases} \tag{6.1}$$

Since $s_1 = 1$, these lengths satisfy

$$\sum_{q=1}^{Q} \ell_q^{(Q)}(\sigma) = 1. \tag{6.2}$$

Let $0 = \tau_0 < \tau_1 < \cdots < \tau_Q = 1$ be defined by

$$\tau_q = \sum_{r=1}^{q} \ell_r^{(Q)}(\sigma). \tag{6.3}$$

Write $U_+(q) = \max\{U(q), 0\}$ and $U_-(q) = \max\{-U(q), 0\}$. Define the nonnegative, piecewise constant strains

$$g_{\pm,Q,\sigma}(t) = U_\pm(q), \qquad \tau_{q-1} \le t < \tau_q. \tag{6.4}$$

Definition 6.1 (Paired CMZ-type channels)**.** Let $X_{\pm,Q,\sigma}$ solve

$$\dot{X}_{\pm,Q,\sigma}(t) = g_{\pm,Q,\sigma}(t) X_{\pm,Q,\sigma}(t), \qquad X_{\pm,Q,\sigma}(0) = 1. \tag{6.5}$$

Their relative terminal deformation is

$$\mathscr{C}_Q(\sigma) = \frac{X_{+,Q,\sigma}(1)}{X_{-,Q,\sigma}(1)}. \tag{6.6}$$

Both amplitudes in (6.5) remain positive. The arithmetic sign occurs only through their ratio.

Proposition 6.2 (Exact arithmetic readout)**.** *For every $Q \ge 1$ and every real $\sigma > 0$,*

$$\log \mathscr{C}_Q(\sigma) = \delta_Q(\sigma). \tag{6.7}$$

Proof. Solving the scalar equations gives

$$\log X_{\pm,Q,\sigma}(1) = \sum_{q=1}^{Q} U_\pm(q) \ell_q^{(Q)}(\sigma). \tag{6.8}$$

Subtracting the two identities in (6.8) yields

$$\log \mathscr{C}_Q(\sigma) = \sum_{q=1}^{Q} U(q) \ell_q^{(Q)}(\sigma). \tag{6.9}$$

Insert (6.1). The result is precisely the finite Abel identity (4.11). □

Remark 6.3 (Why real σ is used). For nonreal s, the differences $s_q^{-s} - s_{q+1}^{-s}$ are complex and cannot serve as positive time lengths. Proposition 6.2 therefore uses real σ. This loses no information in the main equivalence, because convergence of $\delta(\sigma)$ for every real $\sigma > 3/2$ is already equivalent to RH. Complex continuation is treated separately in Section 8.

7. The CMZ deformation equivalence with RH

We now state the exact bridge. It is a theorem about the explicitly defined deformation in Definition 6.1; it does not assume a PDE realisation.

Theorem 7.1 (Ordinal-prime CMZ deformation criterion)**.** *The following statements are equivalent.*

(i) *The Riemann hypothesis holds.*
(ii) *For every real $\sigma > 3/2$, the sequence $\delta_Q(\sigma)$ converges as $Q \to \infty$.*
(iii) *For every real $\sigma > 3/2$, the positive relative deformation $\mathscr{C}_Q(\sigma)$ converges to a finite nonzero limit.*
(iv) *For every real $\sigma > 3/2$,*

$$\lim_{X\to\infty} \sup_{Y \ge X} \left| \log \frac{\mathscr{C}_Y(\sigma)}{\mathscr{C}_X(\sigma)} \right| = 0. \tag{7.1}$$

Proof. Because the coefficients $c_n = \mathbf{1}_{\mu(n)\neq 0}\mu(n) w_{j(n)}$ vanish at every nonsquarefree integer, the squarefree cutoff $\delta_Q(s) = \sum_{q\le Q} c_{s_q} s_q^{-s}$ is exactly the ordinary partial sum through the integer s_Q. Thus convergence on the squarefree clock is equivalent to convergence in the original integer ordering; no rearrangement is being made.

Assume first RH. By the Littlewood criterion (1.10), for every $\eta > 0$,

$$M(x) = O_\eta\left(x^{1/2+\eta}\right). \tag{7.2}$$

The block formula (2.18), the prime number theorem, and (7.2) imply

$$W(x) = O_\eta\left(x^{3/2+\eta}\log(2x)\right). \tag{7.3}$$

Fix $\sigma > 3/2$ and choose η with $0 < \eta < \sigma - 3/2$. Abel summation between integer cutoffs $X < Y$ gives

$$\sum_{X<n\le Y} \frac{\mathbf{1}_{\mu(n)\neq 0}\mu(n)w_{j(n)}}{n^\sigma} = W(Y)Y^{-\sigma} - W(X)X^{-\sigma} \tag{7.4}$$

$$+ \sigma\int_X^Y W(x)x^{-\sigma-1}\,dx. \tag{7.5}$$

By (7.3), the endpoint terms tend to zero and the integral tail is dominated by a constant times

$$\int_X^\infty x^{1/2+\eta-\sigma}\log(2x)\,dx, \tag{7.6}$$

which tends to zero because $1/2 + \eta - \sigma < -1$. Thus the original-order series $\delta(\sigma)$ converges, proving (i)⇒(ii).

Conversely, suppose (ii). Fix any $\epsilon > 0$ and take

$$\sigma = \frac{3}{2} + \frac{\epsilon}{2}. \tag{7.7}$$

Convergence of $\sum c_n n^{-\sigma}$, where $c_n = \mathbf{1}_{\mu(n)\neq 0}\mu(n)w_{j(n)}$, permits Kronecker's lemma and gives

$$W(x) = o\left(x^\sigma\right). \tag{7.8}$$

Combine (7.8) with the two-sided asymptotic (2.20). At points where $M(x) \neq 0$ this gives

$$M(x) = o\left(\frac{x^{\sigma-1}}{\log x}\right) = o\left(\frac{x^{1/2+\epsilon/2}}{\log x}\right). \tag{7.9}$$

At points where $M(x) = 0$, the desired bound is automatic. Therefore $M(x) = O_\epsilon(x^{1/2+\epsilon})$. Since $\epsilon > 0$ was arbitrary, (1.10) gives RH. This proves (ii)⇒(i).

Proposition 6.2 gives $\mathscr{C}_Q(\sigma) = \exp(\delta_Q(\sigma))$. Since the real exponential is a homeomorphism from $\mathbb{R}$ to $(0,\infty)$, (ii) is equivalent to (iii). Finally,

$$\log\frac{\mathscr{C}_Y(\sigma)}{\mathscr{C}_X(\sigma)} = \delta_Y(\sigma) - \delta_X(\sigma), \tag{7.10}$$

so the Cauchy criterion proves the equivalence of (ii) and (iv). □

Remark 7.2 (The first RH-equivalent node). Definitions of S, SS, U, and the finite deformation are unconditional. The first statement in the dependency chain that is equivalent to RH is the tail condition (7.1). Consequently, an Euler estimate that implies this condition with an unattenuated arithmetic normalisation is an RH-strength estimate; it cannot be counted as a routine consequence of the existence of a CMZ gluing.

8. The analytic α criterion and singularity audit

Let

$$\mathcal{H} = \{s \in \mathbb{C} : \Re s > 3/2\}. \tag{8.1}$$

Theorem 8.1 (Shifted reciprocal-zeta criterion)**.** *The following are equivalent.*

(i) *RH holds.*
(ii) *B is holomorphic on $\mathcal{H}$.*
(iii) *The function $\alpha - \delta$, initially defined on $\Re s > 2$, admits a holomorphic continuation to $\mathcal{H}$.*

Proof. By (3.5), poles of B in $\mathcal{H}$ are exactly the points $1+\rho$ arising from zeros ρ of ζ with $\Re\rho > 1/2$; the elementary poles listed in Proposition 3.3 lie outside $\mathcal{H}$. Thus (i) is equivalent to (ii), using the functional-equation symmetry of nontrivial zeros.

In $\mathcal{H}$, the function C is holomorphic by Proposition 3.2. Also A is holomorphic there: its possible denominator singularities arise from zeros of $\zeta(2s)$, which have $\Re s < 1/2$, while the pole at $s = 1$ is outside $\mathcal{H}$. On $\Re s > 2$, identity (3.3) gives

$$\alpha(s) - \delta(s) = A(s) + B(s) + C(s). \tag{8.2}$$

Therefore the right-hand side extends holomorphically to $\mathcal{H}$ exactly when B does. Uniqueness follows from the identity theorem, proving the equivalence of (ii) and (iii). □

Warning 8.2 (Two distinct routes). Theorem 7.1 concerns original-order convergence of $\delta_Q(\sigma)$ on the real axis. Theorem 8.1 concerns continuation of $\alpha - \delta$. A regrouped continuation of δ does not by itself prove original-order convergence, and holomorphy of δ alone does not cancel a pole of B in the full α.

Corollary 8.3 (Abscissa statement)**.** *Let*

$$\Theta = \sup_{\zeta(\rho)=0,\ 0<\Re\rho<1} \Re\rho. \tag{8.3}$$

For the original ordering of the Dirichlet series,

$$\sigma_c(\delta) = \sigma_c(\alpha) = 1 + \Theta. \tag{8.4}$$

Proof. The classical growth-abscissa relation for M gives growth exponent Θ [3]. Proposition 2.6 raises that exponent by one for W, so the abscissa of δ is $1+\Theta$. The signed-prime part of SS has the same exponent, whereas the zero row has partial sums $O(x)$; hence α has the same abscissa. This argument is about ordinary ordering and does not assert a natural boundary or a line of poles. □

9. Conditional realisation by genuine CMZ towers

The exact criterion above uses the CMZ logarithmic law, but its prescribed piecewise strain $g_{\pm,Q,\sigma}$ is not automatically generated by a solution of the Euler equation. We isolate the additional statement needed for a genuine PDE realisation.

Definition 9.1 (Admissible CMZ readout)**.** For fixed real $\sigma > 3/2$, an admissible CMZ readout is a family of positive PDE deformation ratios $\widetilde{\mathscr{C}}_Q(\sigma)$ for which

$$\log \widetilde{\mathscr{C}}_Q(\sigma) = \delta_Q(\sigma) + r_Q(\sigma), \tag{9.1}$$

where the realisation error satisfies

$$\lim_{X\to\infty} \sup_{Y\geq X} |r_Y(\sigma) - r_X(\sigma)| = 0. \tag{9.2}$$

The coefficient of δ_Q in (9.1) is exactly one and is independent of Q and of all spatial carrier frequencies.

Theorem 9.2 (Conditional Euler-CMZ transfer)**.** *Assume that an admissible CMZ readout exists for every real $\sigma > 3/2$. Then RH is equivalent to*

$$\lim_{X\to\infty} \sup_{Y\geq X} \left| \log \frac{\widetilde{\mathscr{C}}_Y(\sigma)}{\widetilde{\mathscr{C}}_X(\sigma)} \right| = 0 \quad \textit{for every real } \sigma > 3/2. \tag{9.3}$$

Proof. Subtract (9.1) at Y and X to obtain

$$\log \frac{\widetilde{\mathscr{C}}_Y(\sigma)}{\widetilde{\mathscr{C}}_X(\sigma)} = \delta_Y(\sigma) - \delta_X(\sigma) \tag{9.4}$$

$$+ r_Y(\sigma) - r_X(\sigma). \tag{9.5}$$

Condition (9.2) shows that (9.3) holds exactly when the δ tails satisfy the Cauchy condition. Apply Theorem 7.1. □

Question 9.3 (Paired-reunion problem)**.** Can one construct two genuine CMZ towers from the ordinal pairs $+p_j, -p_j$ so that their principal Biot–Savart strains cancel after both members have appeared, while the transported reunion defects satisfy (9.2) with the exact, frequency-independent normalisation in (9.1)?

Question 9.3 is [OPEN]. Fixed-duration pairs are compatible with local transport estimates, but long pairs can undergo large hyperbolic deformation before reunion. Allowing the carrier scale to absorb $|U(q)|$ does not answer the question: it changes the readout coefficient and can make any finite datum appear small.

9.1. **Three attempted insertions of the ordinal-prime data.** The two-channel model of Section 6 reads $U(q)$ off an already-defined scalar ODE; it does not feed SS into a genuine solution of a forced Euler or IPM equation. Three routes toward such a feeding are recorded here, because each rules out a naive fix and narrows what a correct insertion must do.

(a) Rescaling the amplitude of a single packet. In the forced construction of [5] (its Theorem 5.1) and in its space–time-smooth extension [7] (its Proposition 5.13, eq. (101), and Corollary 4.10), the leading amplitude of a child packet is not a free parameter. In [5], the parent hypothesis pins $\partial_{x_1} u_1(\bar\omega_1)$ to $[C_0/4, C_0]N^{\varepsilon}$ over a window of length $N^{-\varepsilon/2}$, forcing the next scale $M^{1/2} = \exp\left(\Theta(N^{\varepsilon/2})\right)$; in [7], the amplitude $a_{n+1}(t)$ solves an ODE driven by the parent field and is normalised at the terminal time, $a_{n+1}(1) = 1$, which forces the seed value $a_{n+1}(s_{n+1})$ to be exponentially small. Multiplying an already-built exact solution by an external factor changes the force by a term of the same nonlinear order as the self-interaction already accounted for in (5.8), and no estimate in either source covers that modified force. [OPEN, ruled out as stated]: neither source offers a free per-event amplitude dial.

(b) Continuous exponent insertion. In [7], the scale exponent $\beta_{n+1} \in [\beta_n/2, \beta_n]$ of its Proposition 5.13 is a genuine interval, and the threshold Λ_{sep} of its Proposition 4.2 is proved uniform over that whole interval. It is tempting to set $\beta_{n(q)+1}/\beta_{n(q)} = \frac{3}{4}\exp(\kappa\, c_q s_q^{-\sigma}) \in [\frac{1}{2}, 1]$ and read $\delta_Q(\sigma)$ off $\log \beta_{n(Q)+1}$. Checked against the same paper's global assembly rather than against the one-step proposition alone, this fails for a structural reason: the recursive exponent choice that proves global existence there (its eqs. (172)–(174)) never uses an intermediate value of β_{n+1} — at every level it sets β_{n+1} to exactly β_n or exactly $\beta_n/2$, and the argument proving $\beta_n \downarrow 0$ (its Lemma 6.3) uses this finitely-many-halvings dichotomy directly. A continuously varying exponent schedule is therefore outside what the cited global theorem proves, not an unverified special case of it. [OPEN]: encoding δ_Q in β_{n+1} this way needs a new global-iteration theorem for a continuously varying exponent, which is not in the cited sources. It is also, on its own terms, the wrong kind of fix: the same scheme accepts *any* sequence $x_q \to 0$, so proving the extension would encode SS without constraining it.

(c) Insertion into the parent gradient. Writing $P_{n+1} = S_{\mu_{n+1}}\rho^{(n)}$ for the actual parent of the next packet (eq. (107) of [7]), the real amplitude law there is

$$\frac{\dot a_{n+1}}{a_{n+1}} = C_0 \cos\alpha_{n+1}\, e^{\perp}(\alpha_{n+1}) \cdot \nabla P_{n+1}(0,t), \qquad a_{n+1}(1) = 1. \tag{9.6}$$

Asking that

$$\nabla P_{n(q)+1}(0,t_q) = \nabla P^{\mathrm{cascade}}_{n(q)+1}(0,t_q) + \mathbf{v}\, U(q) + E_q \tag{9.7}$$

for a fixed vector $\mathbf{v}$ and a summable error, so that $U(q)$ enters (9.6) through an output of the construction rather than through a hand-chosen parameter, is structurally the right kind of target: ∇P_{n+1} is forced by the sum of earlier packets, not free. It is also compatible with the actual size of the data: $c_q = O(q\log q)$ grows, and the native CMZ regime is one of growing packet amplitudes,

not decaying ones, unlike attempts (a) and (b) above. No construction of the auxiliary packets in (9.7) is given here or in the sources consulted; it is recorded below as Question 9.24 and is [OPEN].

(d) The single-tower amplitude ratio is impossible, not merely unproven. A natural sharpening of (c) proposes reading $U(q)$ directly off a single tower's amplitude ratio,

$$\log \frac{a_{q+1}(1)}{a_{q+1}(s_{q+1})} = c_q s_q^{-\sigma} + e_q. \tag{9.8}$$

This is excluded outright, not merely unverified.

Proposition 9.4 (Sign obstruction for the single-tower amplitude ratio)**.** *For every admissible family in the sense of* [7] *and every q past the threshold of its Lemma 4.9,*

$$\log \frac{a_{q+1}(1)}{a_{q+1}(s_{q+1})} \geq k_g\, \lambda_q^{\beta_q/8} > 0, \qquad k_g = \frac{1}{96\pi} \tag{9.9}$$

(its eq. (102)). On the other hand $c_q = \varepsilon_q w_{\alpha_q}$, with $\varepsilon_q = \mu(s_q) \in \{\pm 1\}$ changing sign infinitely often and $w_j = p_j - j > 0$ for every $j \geq 1$, so $c_q s_q^{-\sigma}$ changes sign infinitely often, unconditionally. Consequently (9.8) *cannot hold with $e_q \to 0$: its left side is always positive while its right side is negative for infinitely many q.*

Proof. Immediate from (9.9) and the sign pattern of μ. □

This does not merely rule out an unverified special case; it contradicts an already-proved lemma of [7]. A two-tower difference (the architecture of Section 6, now built from genuine packets on each side) evades the sign problem, since a difference of two positive quantities has no fixed sign. It does not evade the size problem, treated in Section 9.3.

9.2. **A generalisation of the layered amplitude bound.** The unforced layered construction of [6] states its self-interaction bound (its Lemma 1) only for geometric initial data $x_n(0) = A^n$, with a single ratio $A > 1$ fixed for the whole tower. The following removes that restriction. It is proved here and is not present in the cited source.

Lemma 9.5 (Bounded-ratio generalisation)**.** *Let $(B_n)_{n\geq 0}$ be positive reals with $B_0 = 1$ and*

$$r_n := B_n/B_{n-1} \in (0, \bar{r}] \qquad (n \geq 1), \qquad \bar{r} > 1. \tag{9.10}$$

Suppose $x_n(t) > 0$ solves $\dot{x}_n(t) = x_n(t) \sum_{j<n} a_j(t) x_j(t)$ with $a_j(t) \geq 1$ and $x_n(0) = B_n$. Let $\zeta > 0$ solve $\zeta = \bar{r}(1 - e^{-\zeta})$. Then, for every $n \geq 0$,

$$\int_{-\zeta}^{0} x_n(t)\, dt \leq \zeta. \tag{9.11}$$

Proof. Put $y_n = \ln x_n$ and $Z_n(T) = \int_{-T}^{0} x_n(t)\, dt$. Since $a_n(t) \geq 1$, $\dot{y}_{n+1}(t) \geq \dot{y}_n(t) + e^{y_n(t)}$; integrating from $t \leq 0$ to 0 and exponentiating,

$$x_{n+1}(t) \leq r_{n+1}\, x_n(t) \exp\Big(- \int_t^0 x_n(s)\, ds \Big). \tag{9.12}$$

Integrating (9.12) over $t \in [-T, 0]$ and using $\frac{d}{dt} \exp(-\int_t^0 x_n(s)ds) = x_n(t) \exp(-\int_t^0 x_n(s)ds)$ gives

$$Z_{n+1}(T) \leq r_{n+1}(1 - e^{-Z_n(T)}). \tag{9.13}$$

Since $x_0 \equiv B_0 = 1$, $Z_0(T) = T$; take $T = \zeta$, so $Z_0(\zeta) = \zeta$. If $Z_n(\zeta) \leq \zeta$, then (9.13), $r_{n+1} \leq \bar{r}$, and monotonicity of $x \mapsto 1 - e^{-x}$ give $Z_{n+1}(\zeta) \leq \bar{r}(1 - e^{-\zeta}) = \zeta$. Induction closes. □

Remark 9.6 (Scope of Lemma 9.5). This replaces the single global ratio A by a bounded, otherwise arbitrary sequence of ratios r_n, so in principle a different target can be assigned at each level. It does not by itself insert SS: the construction of [6] is a finite-time blow-up at $t = 0$, driven by $x_n(0) \to \infty$ along the tower, whereas the quantity needed for Theorem 7.1 is convergence, as

$Q \to \infty$, of a Cauchy tail — the opposite regime. Reusing this lemma for the ordinal-prime problem therefore requires either a genuinely different, non-blow-up time parametrisation, or keeping only the self-interaction estimate and discarding the blow-up end state; neither is done here.

9.3. **A quadratic-summability bound, and the size obstruction to a two-tower comparison.** The following unconditional estimate isolates exactly how small the arithmetic target already is, independently of any PDE construction.

Proposition 9.7 (Quadratic summability of the ordinal-prime target)**.** *Let $d_q(\sigma) := c_q s_q^{-\sigma}$. Then $|d_q(\sigma)| \ll_\sigma q^{1-\sigma}\log(2q)$, and*

$$\sum_{q=1}^{\infty} |d_q(\sigma)|^2 < \infty \qquad \text{for every } \sigma > \frac{3}{2}. \tag{9.14}$$

Proof. Since $\alpha_q \le q$ and $p_j \ll j\log(2j)$ (prime number theorem), $|c_q| = w_{\alpha_q} = p_{\alpha_q} - \alpha_q \ll q\log(2q)$. Since $s_q \asymp q$ (density of squarefree integers), $|d_q(\sigma)| \ll q^{1-\sigma}\log(2q)$, and (9.14) converges exactly when $2\sigma - 2 > 1$, i.e. $\sigma > 3/2$. □

This uses only the prime number theorem and squarefree density — no input equivalent to RH — and shows the target sequence is already small on its own terms. The next result shows this smallness works against, not for, a two-tower realisation built from activated CMZ packets.

Proposition 9.8 (Size obstruction for a two-tower comparison)**.** *Let $\mathcal{T}^+, \mathcal{T}^-$ be two admissible families in the sense of* [7]*, each satisfying its global iteration (so $\beta_q^\pm \downarrow 0$ and $x_q^\pm := \beta_q^\pm \log \lambda_q^\pm \to \infty$, its eq. (176)). If a layer is activated in either tower at step q, its amplitude-gain contribution to that tower's log-amplitude is, by* (9.9)*, at least $k_g \exp(x_q^\pm/8) \to \infty$. An insertion scheme in which exactly one tower is activated at each step cannot satisfy*

$$\left| \log \frac{a_{q+1}^+(1)}{a_{q+1}^+(s_{q+1}^+)} - \log \frac{a_{q+1}^-(1)}{a_{q+1}^-(s_{q+1}^-)} \right| = |d_q(\sigma)| + O(\eta_q), \qquad \eta_q \to 0, \tag{9.15}$$

for q past the threshold where the inactive side contributes 0 *and the active side's contribution exceeds any fixed bound, since the left side then diverges while the right side, by Proposition 9.7, tends to* 0*. If instead both towers are activated at every step,* (9.15) *demands that two independently-evolving, individually unbounded quantities agree to within a vanishing relative error at every scale — a synchronisation condition supplied by no bound in* [5, 6, 7].

Remark 9.9 (What this rules out and what it does not)*.* Proposition 9.8 rules out any two-tower realisation built from the *activated-packet* mechanism of [7] — the crossing of its Lemma 4.8 followed by the gain of its Lemma 4.9 — because that mechanism is, by design, an amplifier: every activation is forced to contribute an unboundedly growing amount, the opposite of the boundedly small, freely signed contribution that Proposition 9.7 shows is actually needed. It does not rule out a construction that avoids activation altogether and instead reads $d_q(\sigma)$ off a quantity that already exists in the cascade as an error term, rather than as an amplified response.

Question 9.10 (Natural-residual injection)**.** The cascade of [7] already produces, at each level, force increments and structured defects G_{q+1}, E_j, Q_j (its eqs. (134)–(135), (154)) that are not designed but forced by the mismatch between the truncated parametrix and the exact equation. Is there a choice of the fixed profiles f, χ, W, together with a rule identifying the activation times s_{q+1} with the squarefree times s_q of Section 4, for which one of these already-occurring residual quantities, read at $(0,1)$, equals $\lambda_\sigma\, c_q s_q^{-\sigma} + e_q$ for a fixed nonzero constant λ_σ independent of q and σ, with $\sum_q |e_q| < \infty$, without choosing any function of the construction using the value c_q itself?

Question 9.10 is [OPEN, speculative] and is, on present evidence, the only candidate not excluded by Propositions 9.4 and 9.8: residual terms are not subject to the activation-gain lower bound of

Lemma 4.9, since they are the leftover error of an approximation rather than the leading amplified response. Whether any such residual is large enough to be read out at all, and small enough to match Proposition 9.7's size, is not addressed by any source consulted here. Answering Question 9.10 in the negative — showing every such residual is either too small to read (unconditionally smaller than the estimates already used to prove force smoothness) or itself tied to an amplifying step, reintroducing Proposition 9.8 — would be as informative as answering it in the affirmative.

9.4. **The zero-harmonic route supplied by SS.** The preceding question can be made substantially sharper. The essential observation is not to prescribe the small signed coefficient c_q as a packet amplitude. Instead, one puts every still-unmatched ordinal pair into its own prime harmonic and asks the quadratic self-interaction to count the unmatched bands. This subsection separates the exact finite algebra from the two estimates that remain open.

For $q \geq 0$, write

$$N_+(q) = \#\{r \leq q : \varepsilon_r = 1\}, \qquad N_-(q) = \#\{r \leq q : \varepsilon_r = -1\}. \tag{9.16}$$

The jth positive and the jth negative squarefree occurrences form the jth ordinal pair. Let $\mathcal{O}(q)$ be the set of labels for which exactly one member has occurred by time q, and give such a label the sign $\sigma_j(q)$ of its member already present. Thus

$$\mathcal{O}(q) = \{\min(N_+, N_-) + 1, \dots, \max(N_+, N_-)\}, \qquad \sigma_j(q) = \operatorname{sgn} R(q). \tag{9.17}$$

The set is empty when $R(q) = 0$.

Lemma 9.11 (Open pairs are read exactly from SS)**.** *For every $q \geq 0$,*

$$\#\mathcal{O}(q) = |N_+(q) - N_-(q)| = |R(q)| = |M(s_q)|. \tag{9.18}$$

Moreover, the complete signed harmonic list up to time q is recoverable from SS: at a squarefree integer s_r,

$$S(s_r) = SS(s_r) - SS(s_r - 1) = \varepsilon_r p_{j(s_r)}. \tag{9.19}$$

[DED]

Proof. The N_+ positive occurrences close exactly the first $\min(N_+, N_-)$ labels against the corresponding negative occurrences, and likewise for N_-. Hence the unmatched labels are precisely the interval in (9.17), whose cardinality is $|N_+ - N_-|$. By (4.1), this difference is $R(q) = M(s_q)$. Finally, (9.19) is the defining finite difference of SS together with the squarefree row in the definition of S. □

Define the finite trigonometric polynomial

$$\mathfrak{A}_q(\vartheta) = \sum_{r=1}^{q} \varepsilon_r \cos(p_{j(s_r)}\vartheta) \tag{9.20}$$

$$= \sum_{j \in \mathcal{O}(q)} \sigma_j(q) \cos(p_j \vartheta). \tag{9.21}$$

The second equality is pointwise: the two members of every closed pair cancel before any limiting operation is taken.

Proposition 9.12 (Exact zero-harmonic pair count)**.** *Let*

$$\langle F \rangle_\tau := \frac{1}{2\pi} \int_0^{2\pi} F(\tau)\, d\tau. \tag{9.22}$$

Then

$$\Big\langle \mathfrak{A}_q(\vartheta + \tau)^2 \Big\rangle_\tau = \Pi_0(\mathfrak{A}_q \star \mathfrak{A}_q) = \frac{1}{2}\,|R(q)|. \tag{9.23}$$

Here Π_0 *and* $\star$ *denote the formal array operations of* [7, (128)–(129)]. *The first member of* (9.23) *is intrinsic, whereas the middle member depends on the displayed array presentation.* [DED]

Proof. Distinct labels carry distinct prime frequencies. Therefore

$$\frac{1}{2\pi}\int_0^{2\pi}\cos(p_j(\vartheta+\tau))\cos(p_k(\vartheta+\tau))\,d\tau=\frac{1}{2}\mathbf{1}_{j=k}. \tag{9.24}$$

The signs square to one, so summing (9.24) over the open labels and using Lemma 9.11 proves the formula. The formal array product gives the same coefficient by the product-to-sum identities. □

We next place this identity inside the local IPM packet calculus. Fix a time t, a transported phase X, a smooth coefficient b supported in the coordinate chart, and a carrier λ. Put

$$z_{q,\tau}(x,t):=b(x,t)\mathfrak{A}_q(\lambda X(x,t)+\tau). \tag{9.25}$$

Let

$$m_0(t)=-C_0\cos\alpha(t)e^{\perp}(\alpha(t)) \tag{9.26}$$

be the leading multiplier vector in [7, (115)–(119)], so that bandwise $u^{[0]}[z]=m_0 z$.

Proposition 9.13 (What the leading array term measures)**.** *For the chosen multiband presentation of* (9.25)*,*

$$\mathrm{Ev}_{\lambda,X}\,\Pi_0\big(u^{[0]}[z_{q,\tau}]\cdot\nabla z_{q,\tau}\big)=\frac{|R(q)|}{4}\,m_0(t)\cdot\nabla(b^2). \tag{9.27}$$

[DED]

Proof. Expand $m_0 b\mathfrak{A}_q\cdot\nabla(b\mathfrak{A}_q)$. The phase derivative has zero mean because $\langle\mathfrak{A}_q\mathfrak{A}_q'\rangle_\tau=0$. The coefficient derivative gives

$$(m_0\cdot\nabla b)b\left\langle\mathfrak{A}_q^2\right\rangle_\tau=\frac{|R(q)|}{4}m_0\cdot\nabla(b^2), \tag{9.28}$$

by Proposition 9.12. □

The factor $1/4$ in (9.27) must not be mistaken for the coefficient of the exact weak residual. Indeed, estimating $(u^{[M]}-u^{[0]})[z]\cdot\nabla z$ before integration by parts loses the fast derivative: the velocity gain $\lambda^{-2\beta}$ from [7, (33)] is accompanied by a factor λ from ∇z. Thus a relative $O(\lambda^{-2\beta})$ error does not follow from that direct product estimate.

The remedy is to recombine the exact velocity first and use the divergence form. For clarity, the needed multiplier estimate is stated explicitly.

Definition 9.14 (Uniform leading-response hypothesis)**.** At the fixed time t, the packet family has leading-response constant C_{vel} if, for every $j\in\mathcal{O}(q)$ and every τ,

$$u^T\big[b\cos\big(p_j(\lambda X+\tau)\big)\big]=m_0 b\cos\big(p_j(\lambda X+\tau)\big)+r_{j,\tau},\qquad \|r_{j,\tau}\|_{L^2}\le C_{\mathrm{vel}}\lambda^{-2\beta}\|b\|_{L^2}. \tag{9.29}$$

This is a stated quantitative hypothesis, not an arithmetic assumption. Proposition 3.4 and its exact remainder estimate in [7] give the corresponding coefficient bounds for a fixed normalised profile class and for the permitted finite harmonic range. They do not state (9.29) uniformly over arbitrary b or arbitrarily many prime bands.

Proposition 9.15 (Exact first-residual recombination and coercivity)**.** *Construct the* $j=1$ *defect* $E_{1,q,\tau}$ *from* (9.25) *by applying the array definitions* [7, (133)–(135)] *band by band. Then, for every smooth test function* φ*,*

$$\langle E_{1,q,\tau}\rangle_\tau=\left\langle u^T(z_{q,\tau})\cdot\nabla z_{q,\tau}\right\rangle_\tau, \tag{9.30}$$

$$\langle\langle E_{1,q,\tau},\varphi\rangle\rangle_\tau=-\frac{|R(q)|}{2}\int b^2 m_0\cdot\nabla\varphi\,dx+\mathcal{R}_{q,\lambda}(\varphi), \tag{9.31}$$

where, under Definition 9.14,

$$|\mathcal{R}_{q,\lambda}(\varphi)| \le C_{\mathrm{vel}}\lambda^{-2\beta}|R(q)|\|b\|_{L^2}^2\|\nabla\varphi\|_{L^\infty}. \tag{9.32}$$

The identities are [DED]*; applying them to the multiband seed is an* [ADAPT] *of the published one-band construction.*

Proof. For $j = 1$, formula (133) of the cited paper reduces to $N_1 = u^{[M]}[z]\cdot\nabla z$. Its formula (135) consists of $\mathrm{Ev}(\Pi_0 N_1)$, the exact multiplier remainder against the parent gradient, and $(u^T - u^{[M]})[z]\cdot\nabla z$. The phase average of the parent-gradient term is zero because it is linear in a nonzero τ-harmonic. The phase average of $\mathrm{Ev}(\Pi_0 N_1)$ equals the phase average of the complete N_1, because every nonzero formal harmonic integrates to zero. Adding the last remainder reconstructs $u^T(z)\cdot\nabla z$, proving (9.30).

The IPM velocity $u^T(z)$ is divergence free and the packet is compactly supported inside the chart, so there is no spatial boundary term and

$$\Big\langle\Big\langle u^T(z_{q,\tau})\cdot\nabla z_{q,\tau},\varphi\Big\rangle\Big\rangle_\tau = -\Big\langle\int z_{q,\tau}u^T(z_{q,\tau})\cdot\nabla\varphi\,dx\Big\rangle_\tau. \tag{9.33}$$

Linearity of u^T preserves the external phase harmonic. Hence all cross-band terms vanish under the τ average. The leading response in each surviving diagonal band contributes $-\frac{1}{2}\int b^2 m_0\cdot\nabla\varphi$, independently of its sign. Cauchy–Schwarz and (9.29), summed over the $|R(q)|$ surviving bands, give (9.32). □

Corollary 9.16 (Local readout of the open-pair count)**.** *Assume $m_0(t)\neq 0$. If* $\operatorname{supp} b$ *lies in a coordinate ball, there is a smooth periodic test function φ_t such that $m_0(t)\cdot\nabla\varphi_t = 1$ on* $\operatorname{supp} b$*. Consequently,*

$$\left|\frac{-2}{\|b\|_{L^2}^2}\left\langle\langle E_{1,q,\tau},\varphi_t\rangle\right\rangle_\tau - |R(q)|\right| \le 2C_{\mathrm{vel}}\lambda^{-2\beta}|R(q)|\|\nabla\varphi_t\|_{L^\infty}. \tag{9.34}$$

In particular, the right choice of a sufficiently large carrier gives a finite-stage coercive measurement of $|M(s_q)|$. [DED]

Proof. Choose a cutoff equal to one on $\operatorname{supp} b$ and multiply it there by the affine coordinate $(m_0/|m_0|^2)\cdot x$. Periodise inside the chart and apply Proposition 9.15. Compact support removes every edge term. Notice that this construction is unavailable at a time where $m_0 = 0$; one must work on a subinterval on which $|m_0|$ is bounded away from zero. □

Warning 9.17 (Harmonic capacity is finite at fixed β). In Proposition 3.4 and Corollary 3.6 of [7], the input harmonic satisfies

$$1 \le \ell \le 2M_\beta + 1, \qquad M_\beta = \lceil\beta^{-2}\rceil. \tag{9.35}$$

Therefore the published estimates apply to (9.25) only when

$$H_q := \max_{j\in\mathcal{O}(q)} p_j \le 2M_\beta + 1, \tag{9.36}$$

with $H_q = 0$ if the open set is empty. Every fixed q can be accommodated by decreasing β, but this changes the derivative budget, the number of parametrix steps, and the scale thresholds. It supplies no estimate uniform in q.

The complete perturbation residual of the cited construction is

$$E_{q,\tau}^{\mathrm{pert}} = A_{0,q,\tau} + \sum_{j=1}^{M_\beta} E_{j,q,\tau} + Q_{M_\beta,q,\tau}. \tag{9.37}$$

The phase averages of the linear activation term and of the terminal nonzero-harmonic defect vanish. However, the later deposited zero modes E_j, $j \geq 2$, need not vanish and can in principle cancel the coercive E_1 contribution. This is the first genuine wall.

Define the carrier-normalised observable

$$\mathfrak{Z}_q(t) := -\frac{2}{\|b(\cdot,t)\|_{L^2}^2} \left\langle \left\langle E_{q,\tau}^{\mathrm{pert}}(\cdot,t), \varphi_t \right\rangle \right\rangle_\tau . \tag{9.38}$$

The normalisation is essential: an unnormalised estimate for E^{pert} may be small only because the seed amplitude and its support shrink with λ.

Theorem 9.18 (Conditional zero-harmonic implication to RH)**.** *Suppose that for every sufficiently large q one can choose admissible CMZ data satisfying* (9.36)*, with constants independent of q, such that*

$$c\,|R(q)| \leq |\mathfrak{Z}_q(t_q)|, \qquad c > 0, \tag{9.39}$$

$$|\mathfrak{Z}_q(t_q)| \leq C_\eta q^{1/2+\eta} \qquad \text{for every } \eta > 0. \tag{9.40}$$

If (9.39) *is proved by controlling the later CMZ corrections, and* (9.40) *is proved by a fluid estimate that does not assume a Mertens bound or an equivalent form of RH, then RH follows.* [DED, conditional]

Proof. Combining the two displayed inequalities gives $|R(q)| = O_\eta(q^{1/2+\eta})$. Since $R(q) = M(s_q)$ and the squarefree integers have positive density, $s_q \asymp q$. Between two consecutive squarefree integers the Mertens function is constant. Hence

$$M(x) = O_\eta(x^{1/2+\eta}) \tag{9.41}$$

for every $\eta > 0$, and Littlewood's criterion implies RH. □

Question 9.19 (The precise CMZ–RH residual problem)**.** Can one prove (9.39) and (9.40) with a common q-uniform admissible family, despite the growing prime-harmonic capacity and shrinking seed normalisation? [OPEN]

This question is the concrete continuation of Question 9.10. Its first half asks for triangular or sign-definite control of the later zero-mode deposits; its second half asks for a genuinely PDE upper bound. The first-residual identity (9.31) is the promised nontrivial grip on CMZ, but neither half of Question 9.19 is a theorem of the four cited fluid papers.

9.5. **Audit of the complementary depth-transform proposal.** There is a second residual encoding, based on the depth $h_j = |d_j|$ of the jth ordinal pair rather than on the number of pairs open at one squarefree time. Part of this proposal is exact, but its promotion from an affine model to a complete ABC child is not. We record both facts so that the valid algebra can be reused without silently assuming the transfer.

Put

$$n_h(J) = \#\{j \leq J : |d_j| = h\}, \qquad N_J(w) = \sum_{h \geq 1} n_h(J) w^h. \tag{9.42}$$

This is a polynomial for every fixed J. In the affine periodic IPM model, let $K_{r,h}$ be the first-deposit weight from depth h to zero after r nearest-neighbour steps, and define

$$\mathcal{A}_r(J) = \sum_{h \leq r} K_{r,h} n_h(J), \qquad A_J(z) = \sum_{r \geq 1} \mathcal{A}_r(J) z^r. \tag{9.43}$$

In the high-frequency limit the one-depth generating function is

$$F_h(z) = \xi^h \left(1 + h\frac{1+\xi^2}{1-\xi^2}\right), \qquad z = \frac{\xi}{1-\xi^2}. \tag{9.44}$$

Proposition 9.20 (Closed transform between deposits and depths)**.** *Under the affine recurrence* (9.44)*,*

$$A_J\Big(\frac{\xi}{1-\xi^2}\Big) = N_J(\xi) + \frac{1+\xi^2}{1-\xi^2}\,\xi N_J'(\xi), \tag{9.45}$$

$$N_J(\xi) = \frac{1+\xi^2}{\xi}\int_0^{\xi} \frac{1-t^2}{(1+t^2)^2} A_J\Big(\frac{t}{1-t^2}\Big)\,dt. \tag{9.46}$$

Both identities hold as formal power-series identities, coefficient by coefficient. In particular,

$$\mathsf{S}_M(J) := J - \sum_{h=1}^{M} [\xi^h] N_J(\xi) = \#\{j \le J : |d_j| > M\}. \tag{9.47}$$

[DED, affine model]

Proof. Because N_J is a polynomial, summing (9.44) over h causes no infinite-sum interchange:

$$\sum_h n_h \xi^h = N_J(\xi), \qquad \sum_h h n_h \xi^h = \xi N_J'(\xi). \tag{9.48}$$

This proves (9.45). Rearrangement gives the linear equation

$$N_J'(\xi) + \frac{1-\xi^2}{\xi(1+\xi^2)} N_J(\xi) = \frac{1-\xi^2}{\xi(1+\xi^2)} A_J\Big(\frac{\xi}{1-\xi^2}\Big). \tag{9.49}$$

Its integrating factor is $\xi/(1+\xi^2)$ and $N_J(0) = 0$, which yields (9.46). Formula (9.47) is then the definition of the depth counts. □

At finite affine frequency, take

$$a_h(\lambda,\nu) = 1 + \frac{h\lambda^2}{h^2\lambda^2+\nu^2}, \qquad c_h(\lambda,\nu) = 1 - \frac{h\lambda^2}{h^2\lambda^2+\nu^2}, \tag{9.50}$$

and let

$$F_h^{(\lambda,\nu)}(z) = z\{a_h F_{h-1}^{(\lambda,\nu)} - c_h F_{h+1}^{(\lambda,\nu)}\}, \qquad F_0^{(\lambda,\nu)} = 1, \tag{9.51}$$

with $F_h^{(\lambda,\nu)} = O(z^h)$.

Proposition 9.21 (Finite-frequency triangular inversion)**.** *For $1 \le h \le M$,*

$$K_{h,h}(\lambda,\nu) = [z^h] F_h^{(\lambda,\nu)} = \prod_{j=1}^{h} a_j(\lambda,\nu) > 0. \tag{9.52}$$

Consequently the finite deposit matrix $\mathbb{K}_M = (K_{r,h})_{1\le h\le r\le M}$ is triangular and invertible, with

$$\det \mathbb{K}_M = \prod_{h=1}^{M}\prod_{j=1}^{h} a_j(\lambda,\nu) > 0. \tag{9.53}$$

No limit $\lambda \to \infty$ is needed. [DED, affine model]

Proof. To reach zero from h in exactly h nearest-neighbour steps there is only the strictly descending path. Its weight is the product in (9.52). Also $a_j > 1$ for $\lambda, \nu > 0$. Triangularity and the determinant formula follow. □

The exact part that does survive inside the published ABC arrays is the formal tangent recursion. Freeze the parent P, the phase X, the multiplier expansions, and every propagator of one

already constructed child, and perturb only its seed: $z_1^\eta = z_1 + \eta Z$. With dots denoting $\partial_\eta|_{\eta=0}$, differentiation of (133)–(136) gives

$$\dot N_r = u^{[M]}[\dot z_r]\cdot\nabla T_r + u^{[M]}[z_r]\cdot\nabla\dot T_r + u^{[M]}[\dot T_{r-1}]\cdot\nabla z_r + u^{[M]}[T_{r-1}]\cdot\nabla\dot z_r, \tag{9.54}$$

$$\dot Q_r = \Pi_{>0}\dot N_r + (u^{[M]} - u^{[0]})[\dot z_r]\cdot\nabla P, \tag{9.55}$$

$$\dot z_{r+1} = -\mathcal{I}_r\dot Q_r, \tag{9.56}$$

where $\mathcal{I}_r$ denotes the bandwise Duhamel operator used in (136). These are exact finite differentiations. They do not differentiate the activation time, parent, transported phase, or global iteration.

Warning 9.22 (The proposed ABC transfer is not yet a theorem). The estimates for z_r and Q_r in Lemmas 5.7–5.9 of [7] are upper bounds. They do not imply, after normalising a depth-r deposit, the operator expansion

$$\mathscr{L}_r^{\mathrm{ABC}} = \mathscr{L}_{\mathrm{unit}} + O_{\beta,r}(L^{C_{\beta,r}}\lambda^{-\beta}), \tag{9.57}$$

nor the matrix convergence

$$\mathbb{K}_M^{\mathrm{ABC}}(\lambda) = \mathbb{K}_M^{\infty} + O(L^{C_{\beta,M}}\lambda^{-\beta}). \tag{9.58}$$

Upper bounds alone provide neither a lower normalisation for z_1 nor relative control after repeated Duhamel steps. Moreover the real seed is localised and transported, the coefficients are variable, the interactions with $z_2,\dots,z_r$ remain in (9.54), and the exact multiplier remainders and the scalar functional extracting each deposit must all be controlled. The constants must then be followed simultaneously up to $M = \lceil\beta^{-2}\rceil$.

Thus (9.57) is an [ADAPT, OPEN] transfer lemma, not a consequence already proved by ABC. Continuity of the determinant would give invertibility of $\mathbb{K}_M^{\mathrm{ABC}}$ only after this lemma is established. Even that invertibility would merely recover the depth data deliberately placed in the seed; an independent PDE estimate forcing $\mathsf{S}_M(J) = 0$ at $M \asymp J^{1/2+\eta}$ would still be required to obtain RH. Therefore the claimed "native ABC equivalence with RH" is, at present, a conditional encoding and is not incorporated as a theorem.

Warning 9.23 (What a completed insertion would and would not show about RH). Suppose Question 9.24 were answered completely: a genuine, forced, jointly smooth family of Euler or IPM solutions were built whose real deformation ratio equals $\exp(\delta_Q(\sigma))$ exactly, for every $\sigma > 3/2$. By Theorem 9.2, this would upgrade statement (iv) of Theorem 7.1 from a toy-ODE statement to a genuine PDE statement, closing Question 9.3. It would *not* decide the Riemann hypothesis, and would not bring a decision closer, for a specific reason: the equivalence of RH with convergence of $\delta_Q(\sigma)$ is already proved unconditionally in Theorem 7.1 by Littlewood's criterion alone, with no reference to any PDE. A PDE realisation re-encodes $\delta_Q(\sigma)$; it supplies no independent information about $M(x)$. Whether $\delta_Q(\sigma)$ converges would remain exactly as open after the realisation as before it. Propositions 9.4 and 9.8 illustrate the same point concretely: the two most natural native candidates fail not from absence of a proof but from outright conflict with already-proved lemmas of the cited sources, which is itself evidence that the activated-packet mechanism is the wrong tool for producing a small, freely signed quantity — Tao's own description of the method (blog post of 7 September 2026) is that it is designed to locate an "exploitable instability" in a background flow, which is definitionally the opposite goal from bounding how much instability a construction can produce.

The only way a construction of this kind could contribute to deciding RH is if the completed PDE side supplied an inequality that is not derived by first estimating c_q or $U(q)$ arithmetically and then importing the estimate into the packets, but is instead a bound holding for every admissible family of auxiliary packets of the type in (9.7), proved by genuinely fluid-dynamical means (an energy identity, a monotonicity formula, a maximum principle for the transported angle or amplitude), and that, when specialised to the ordinal-prime packets, gives a bound on $M(x)$ at least as strong

as $O_\eta(x^{1/2+\eta})$. No such universal bound exists in the literature consulted here: the multiscale method of [5, 6, 7, 8] proves *existence* of a blow-up for a chosen forcing, which is the opposite kind of statement from a *universal* bound holding for every admissible forcing. [OPEN, speculative]: whether any fluid-mechanical monotonicity of this kind exists for the CMZ mechanism is not addressed by the present manuscript. Absent such a bound, every result in this section, however completed, is a physical model of the Mertens problem and not a route to its resolution.

Question 9.24 (Parent-insertion problem)**.** Can auxiliary packets, attached to the cascade of [7] at the activation times of the ordinal pairs, be constructed so that (9.7) holds with $\sum_q |E_q|\,(s_q^{-\sigma} - s_{q+1}^{-\sigma}) < \infty$, while the resulting force increments remain summable in the sense used in Proposition 6.2?

Question 9.24 is [OPEN] and replaces Question 9.3 as the concrete remaining problem; per Warning 9.23, closing it would settle Question 9.3 without settling RH. Propositions 9.4 and 9.8 show that the natural ways of specialising Question 9.24 to the activated-packet mechanism fail outright; Question 9.10 is the specific remaining instance not excluded by either.

10. Boundary, convergence, and interchange audit

Proposition 10.1 (Complete audit of the logical dependencies)**.** *The proof has the following dependency chain:*

$$(S, SS) \longrightarrow U(q) \longrightarrow \delta_Q(\sigma) \longrightarrow \mathscr{C}_Q(\sigma) \longrightarrow \textit{tail stability} \longrightarrow \mathrm{RH}. \tag{10.1}$$

The first three arrows are finite identities. The fourth is the exact scalar ODE solution. The final arrow uses the original-order convergence theorem and the Littlewood criterion. No step before tail stability assumes RH. The zero-harmonic continuation has a second, presently conditional chain:

$$SS \longrightarrow \mathcal{O}(q) \longrightarrow \Pi_0(\mathfrak{A}_q \star \mathfrak{A}_q) \longrightarrow \langle E_{1,q,\tau}\rangle_\tau \dashrightarrow \mathfrak{Z}_q \dashrightarrow |M(s_q)| \ll_\eta q^{1/2+\eta} \longrightarrow \mathrm{RH}. \tag{10.2}$$

The solid arrows are proved in Lemmas 9.11–9.15; the dashed arrows are exactly the two open estimates (9.39)–(9.40).

Proof. The first arrow is Proposition 4.2; the second is Lemma 4.3; the third is Proposition 6.2; and the last two arrows are Theorem 7.1. □

The analytic and finite operations used above have the following status.

A1. All rearrangements defining $A + B + C + \delta$ take place first in $\Re s > 2$, where coefficient bounds give absolute convergence.
A2. Formula (4.11) is finite and valid for every complex s; it requires no convergence theorem.
A3. The physical CMZ clock uses real $\sigma > 0$, ensuring positivity of all interval lengths.
A4. The unfinished-pair boundary is retained as $U(Q)s_Q^{-\sigma}$. Under RH it vanishes by (7.3); in the converse direction it is part of δ_Q and is not separately discarded.
A5. The integral defining C is holomorphic in $\Re s > 3/2$ by the explicit majorant (3.13). No continuation of δ is used in that argument.
A6. Singularities of B are not promoted to singularities of α. The analytic criterion concerns the proved continuation of $A + B + C = \alpha - \delta$.
A7. The CMZ literature estimates concern prescribed PDE layers. They do not establish Definition 9.1; that definition is an additional hypothesis in Theorem 9.2.
A8. The phase average in (9.22) is over a finite trigonometric polynomial, so all exchanges with the spatial integral and the finite harmonic sum follow from Fubini without convergence issues.
A9. In (9.33), the packet is compactly supported inside the coordinate chart and the ambient IPM velocity is periodic and divergence free. Thus there is neither a chart-boundary contribution nor a torus-boundary contribution.

A10. The test function in Corollary 9.16 is used only when $m_0(t) \neq 0$. Its gradient norm is part of the error constant; no uniform estimate is claimed through a zero of $\cos\alpha(t)$.

A11. The formal Π_0 is presentation-dependent. Every asserted physical readout is therefore expressed through the intrinsic phase average (9.22), not through an identification of Π_0 with a spatial Fourier projection.

A12. At fixed β, the permitted harmonic range is finite. The passage $q \to \infty$ is not justified by choosing $\beta = \beta(q)$ unless all profile, remainder, support, and derivative constants are proved uniform in that joint limit.

Remark 10.2 (A failed shortcut). It is insufficient to prove merely that each finite arithmetic profile can be inserted into a CMZ layer after choosing an arbitrarily large carrier scale. Such a choice may make the PDE residual small while the normalised logarithmic deformation still grows with Q. The uniform, unattenuated identity (9.1) and the tail estimate (9.2) are the properties that discriminate the RH problem from finite gluing flexibility.

11. Conclusion and research route

The ordinal-prime transform produces an exact open-pair strain $U(q)$ that is readable from SS. Finite Abel summation turns this strain into the residual Dirichlet polynomial δ_Q. Splitting U into positive and negative parts permits two positive logarithmic-stretching channels, and their relative deformation equals $\exp(\delta_Q)$. Theorem 7.1 is therefore an exact dynamical reformulation of RH, with the physical clock confined to the fixed interval $[0, 1]$.

The zero-harmonic continuation is more specific than the former parent-insertion question. It uses SS to build the open-band polynomial $\mathfrak{A}_q$, for which the quadratic phase mean is exactly $|M(s_q)|/2$. It then identifies this quantity inside the first exact CMZ residual after the multiplier remainder is recombined and the derivative is moved to a test function. This is a genuine finite-stage CMZ mechanism, not the scalar deformation model of the earlier sections.

It is not yet a proof of RH. The two walls are now explicit: later deposited zero modes must not cancel the E_1 coercivity, and the full residual divided by the shrinking seed energy must satisfy a $q^{1/2+\eta}$ upper bound by an independent fluid argument. In addition, the prime harmonics must fit through the finite capacity $2\lceil\beta^{-2}\rceil + 1$ with uniform constants. These requirements are collected in Question 9.19; if they hold, Theorem 9.18 proves RH. No cited CMZ theorem currently gives them.

The route count is as follows: four [LIT] multiscale inputs, one [ADAPT] two-channel stretching model, two principal [DED] bridges (finite readout and RH equivalence), one [DED] generalisation of an existing lemma (Lemma 9.5), one [DED] unconditional summability bound (Proposition 9.7), four examined insertion attempts of which two are excluded outright by direct conflict with already-proved lemmas (Propositions 9.4, 9.8) and two remain unproved extensions, one exact [DED] open-pair zero-harmonic identity, one [ADAPT] exact first-residual recombination, and one explicitly [OPEN] normalised-residual problem (Question 9.19). Unlike a mere realisation of the old scalar readout, the latter would imply RH only because it demands the new independent upper estimate (9.40). The complementary depth proposal contributes two exact affine-model inversions and the exact frozen-child tangent equations, but its normalised transfer to a complete ABC child remains [ADAPT, OPEN].

Notebook History

Version 1.0: definitions of ζ, μ, λ, and Λ; fundamental properties of S, SS, and α; audited CMZ proof summary; finite-boundary readout; full CMZ deformation equivalence; analytic singularity audit; and conditional genuine-PDE transfer.

Version 2.0: examined three candidate insertions of the ordinal-prime data into a genuine forced Euler/IPM tower (amplitude rescaling, continuous exponent encoding, parent-gradient insertion), and showed each is either an unproved extension of the cited theorems or an encoding that does not

constrain the arithmetic input; proved a bounded-ratio generalisation of the layered self-interaction bound of Córdoba, Martínez-Zoroa, and Zheng; replaced the paired-reunion question by the parent-insertion problem; and recorded, as a standalone warning, the precise sense in which completing any of these routes would not by itself decide the Riemann hypothesis.

Version 3.0: proved that reading the ordinal-prime data off a single tower's amplitude-gain ratio is impossible, not merely unproved, by direct conflict with the amplitude-gain lemma of Alpöge, Buckmaster, and Coiculescu; proved an unconditional quadratic-summability bound for the ordinal-prime target sequence; used it to show a two-tower comparison of activated packets is excluded for the same structural reason, restated precisely; and isolated residual, non-amplified terms of the construction as the one candidate excluded by neither obstruction, posed as the natural-residual injection question.

Version 4.0: inserted the open ordinal pairs encoded by SS into distinct prime harmonics; proved their exact zero-harmonic energy $|M(s_q)|/2$; distinguished the formal array projection from an intrinsic global phase average; recombined the first structured defect with the exact multiplier remainder; proved the resulting weak coercive readout, including support and boundary terms; corrected the tempting but invalid direct $\lambda^{-2\beta}$ remainder count; recorded the finite harmonic-capacity condition; and isolated the two genuinely open uniform estimates whose joint proof would imply RH. It also audited the proposed depth continuation: incorporated the correct closed generating transform, finite-frequency triangular inversion, and exact frozen-child linearisation, while separating the unproved normalised ABC transfer from those identities.

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